

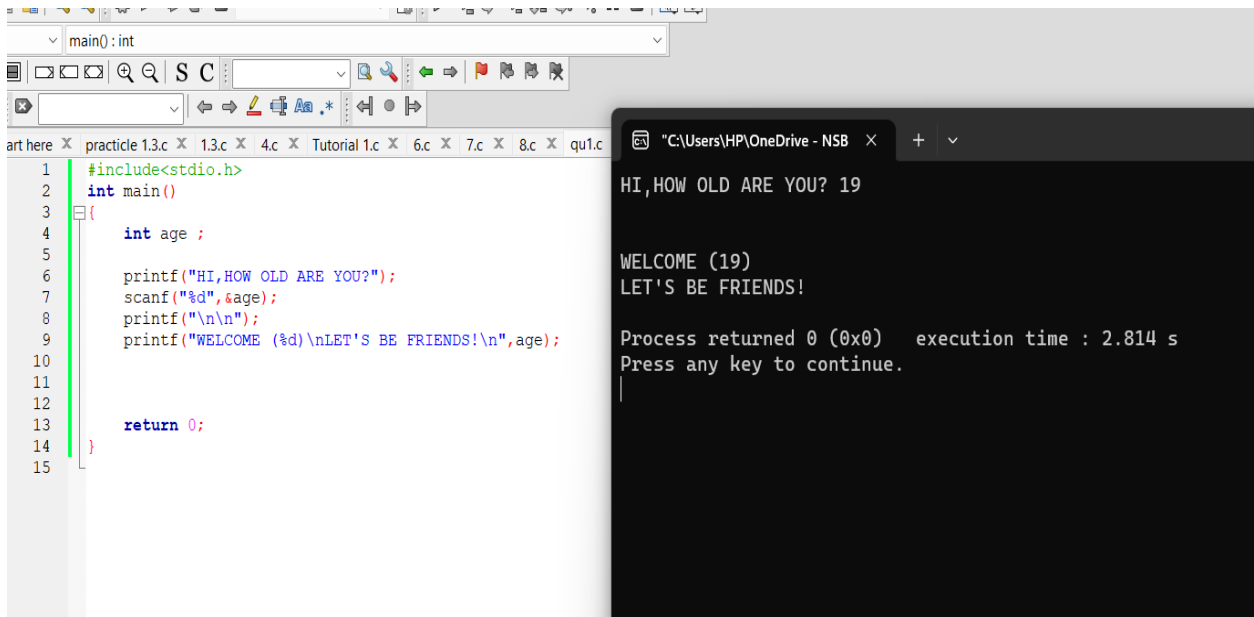
PRACTICAL - 2

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Degree Program Name : Technology Management

```
1. #include<stdio.h>
   int main()
   {
       int age ;

       printf("HI,HOW OLD ARE YOU?");
       scanf("%d",&age);
       printf("\n\n");
       printf("WELCOME (%d)\nLET'S BE FRIENDS!\n",age);

       return 0;
   }
```



The screenshot shows a C program being compiled and executed. The code is displayed in a text editor window on the left, and the output is shown in a terminal window on the right.

Code in the text editor:

```
1 #include<stdio.h>
2 int main()
3 {
4     int age ;
5
6     printf("HI,HOW OLD ARE YOU?");
7     scanf("%d",&age);
8     printf("\n\n");
9     printf("WELCOME (%d)\nLET'S BE FRIENDS!\n",age);
10
11
12
13     return 0;
14 }
15
```

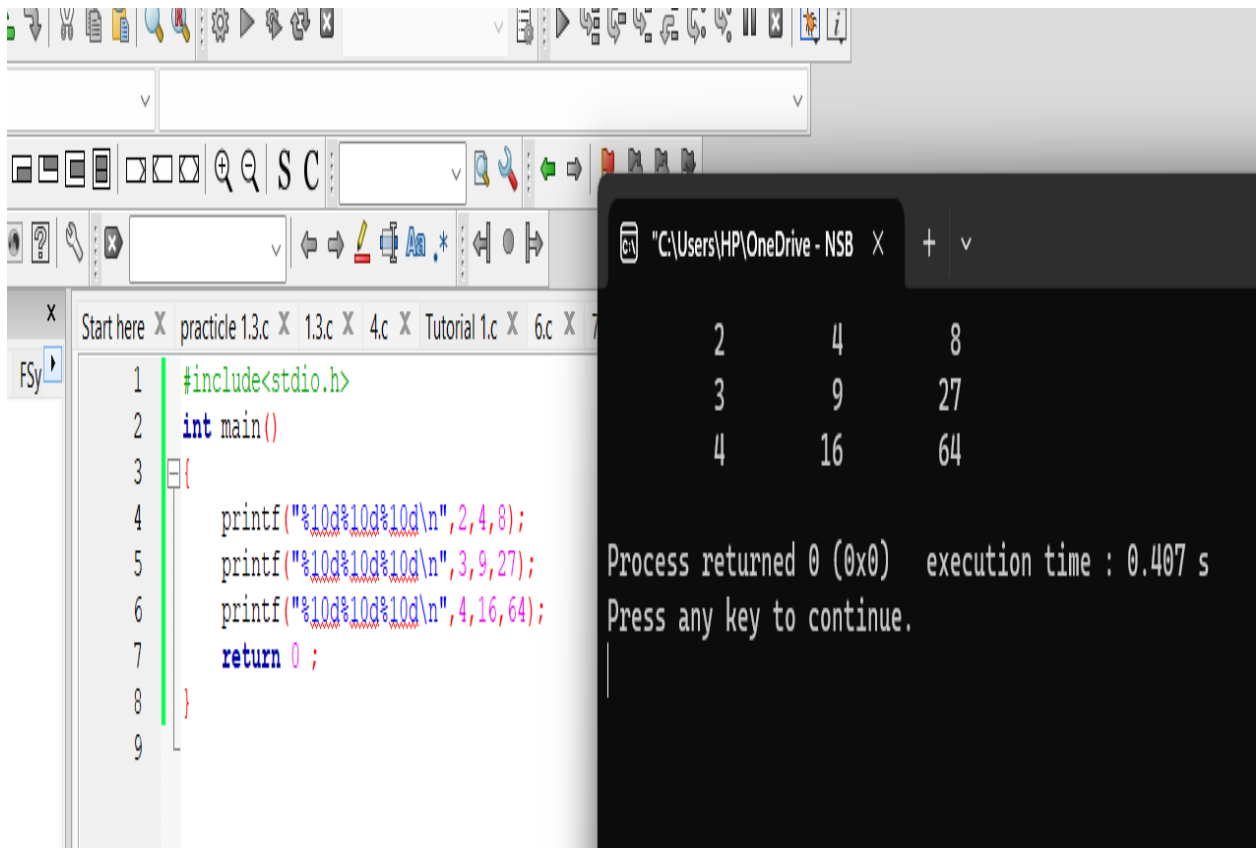
Output in the terminal window:

```
HI,HOW OLD ARE YOU? 19

WELCOME (19)
LET'S BE FRIENDS!

Process returned 0 (0x0)   execution time : 2.814 s
Press any key to continue.
```

```
2. #include<stdio.h>
   int main()
   {
       printf("%10d%10d%10d\n",2,4,8);
       printf("%10d%10d%10d\n",3,9,27);
       printf("%10d%10d%10d\n",4,16,64);
       return 0 ;
   }
```



The screenshot shows a code editor window with the following C code:

```
1 #include<stdio.h>
2 int main()
3 {
4     printf("%10d%10d%10d\n",2,4,8);
5     printf("%10d%10d%10d\n",3,9,27);
6     printf("%10d%10d%10d\n",4,16,64);
7     return 0 ;
8 }
9
```

Below the code editor, a terminal window is open, displaying the output of the program:

```
2         4         8
3         9        27
4        16        64
```

Below the output, the terminal shows the process return status and execution time:

```
Process returned 0 (0x0)   execution time : 0.407 s
Press any key to continue.
```

```

3. #include <stdio.h>
   int main()
   {
       int Dist,Time,Avgspeed;;

       printf("Enter Distance Travelled:");
       scanf("%d",&Dist);

       printf("Enter the Time :");
       scanf("%d",&Time);

       Avgspeed = (Dist/Time);
       printf("Average Speed is %d ",Avgspeed);
       return 0;
   }

```

The screenshot shows a C program being executed in a Windows environment. The code defines three integer variables: `Dist`, `Time`, and `Avgspeed`. It prompts the user to enter distance and time, then calculates the average speed using integer division (`Dist/Time`). The output shows that for an input distance of 1000 and time of 2.4, the average speed is calculated as 500, which is incorrect due to integer truncation.

```

1  #include <stdio.h>
2  int main()
3  {
4      int Dist,Time,Avgspeed;;
5
6      printf("Enter Distance Travelled:");
7      scanf("%d",&Dist);
8
9      printf("Enter the Time :");
10     scanf("%d",&Time);
11
12     Avgspeed = (Dist/Time);
13     printf("Average Speed is %d ",Avgspeed);
14     return 0;
15 }

```

Output:

```

Enter Distance Travelled: 1000
Enter the Time : 2.4
Average Speed is 500
Process returned 0 (0x0)   execution time : 12.702 s
Press any key to continue.

```

1 What would be the problem? Why? How to fix the problem?

- When we input integer numbers to all the variables then we can get only integer value as a output. To solve this problem, we should use the Float data type then we can get decimals in the output. As I mentioned in below ,

```

#include <stdio.h>
int main()
{
    int Dist;
    float Time,Avgspeed;

    printf("Enter Distance Travelled:");
    scanf("%d",&Dist);

    printf("Enter the Time :");
    scanf("%f",&Time);

    Avgspeed = (Dist/Time);
    printf("Average Speed is %f ",Avgspeed);
    return 0;
}

```

```

practicle 1.3.c 1.3.c X 4.c X Tutorial 1.c X 6.c X 7.c X *8.c
1  #include <stdio.h>
2  int main()
3  {
4      int Dist;
5      float Time,Avgspeed;
6
7      printf("Enter Distance Travelled:");
8      scanf("%d",&Dist);
9
10     printf("Enter the Time :");
11     scanf("%f",&Time);
12
13     Avgspeed = (Dist/Time);
14     printf("Average Speed is %f ",Avgspeed);
15     return 0;
16 }
17

```

```

"C:\Users\HP\OneDrive - NSB X + v
Enter Distance Travelled: 1000
Enter the Time :2.15
Average Speed is 465.116272
Process returned 0 (0x0)   execution time : 11.019 s
Press any key to continue.

```

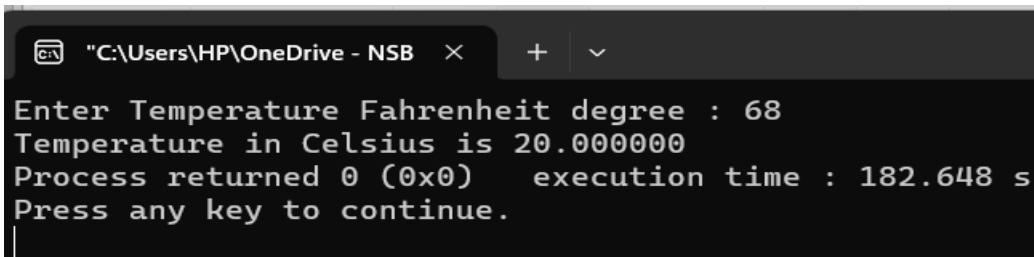
```
4 . #include<stdio.h>
   int main()
   {
       float cel,fah;

       printf("Enter Temperature Fahrenheit degree :");
       scanf("%f",&fah);

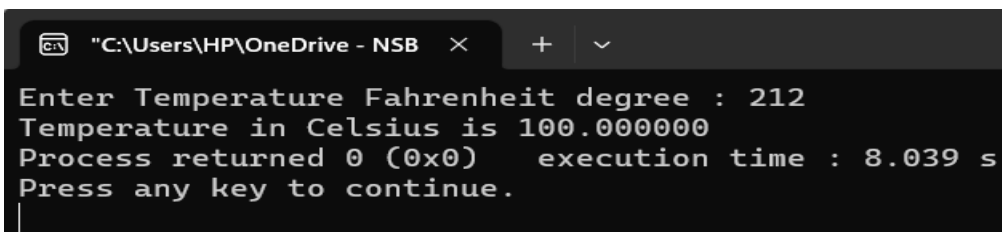
       cel = (5.0/9.0)*(fah-32);

       printf("Temperature in Celsius is %f",cel);

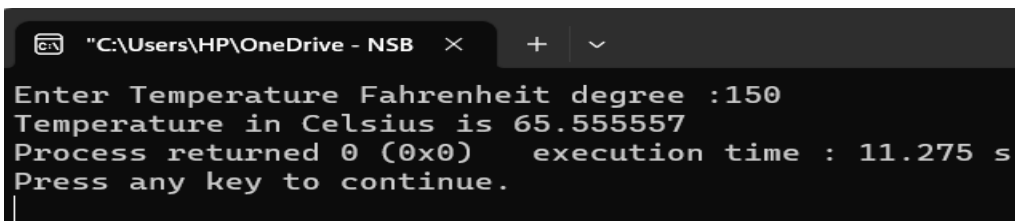
       return 0;
   }
```



```
"C:\Users\HP\OneDrive - NSB"
Enter Temperature Fahrenheit degree : 68
Temperature in Celsius is 20.000000
Process returned 0 (0x0) execution time : 182.648 s
Press any key to continue.
```



```
"C:\Users\HP\OneDrive - NSB"
Enter Temperature Fahrenheit degree : 212
Temperature in Celsius is 100.000000
Process returned 0 (0x0) execution time : 8.039 s
Press any key to continue.
```



```
"C:\Users\HP\OneDrive - NSB"
Enter Temperature Fahrenheit degree :150
Temperature in Celsius is 65.555557
Process returned 0 (0x0) execution time : 11.275 s
Press any key to continue.
```

```
"C:\Users\HP\OneDrive - NSB  × + ∨  
Enter Temperature Fahrenheit degree : 0  
Temperature in Celsius is -17.777779  
Process returned 0 (0x0)  execution time : 4.625 s  
Press any key to continue.
```

```
"C:\Users\HP\OneDrive - NSB  × + ∨  
Enter Temperature Fahrenheit degree : 212  
Temperature in Celsius is 100.000000  
Process returned 0 (0x0)  execution time : 8.0  
Press any key to continue.  
|
```

```
"C:\Users\HP\OneDrive - NSB  × + ∨  
Enter Temperature Fahrenheit degree :-22  
Temperature in Celsius is -30.000000  
Process returned 0 (0x0)  execution time : 3.957 s  
Press any key to continue.  
|
```

```
"C:\Users\HP\OneDrive - NSB  × + ∨  
Enter Temperature Fahrenheit degree : -200  
Temperature in Celsius is -128.888885  
Process returned 0 (0x0)  execution time : 7.479 s  
Press any key to continue.  
|
```